AI IN MEDICAL IMAGING (SINOGRAMS)

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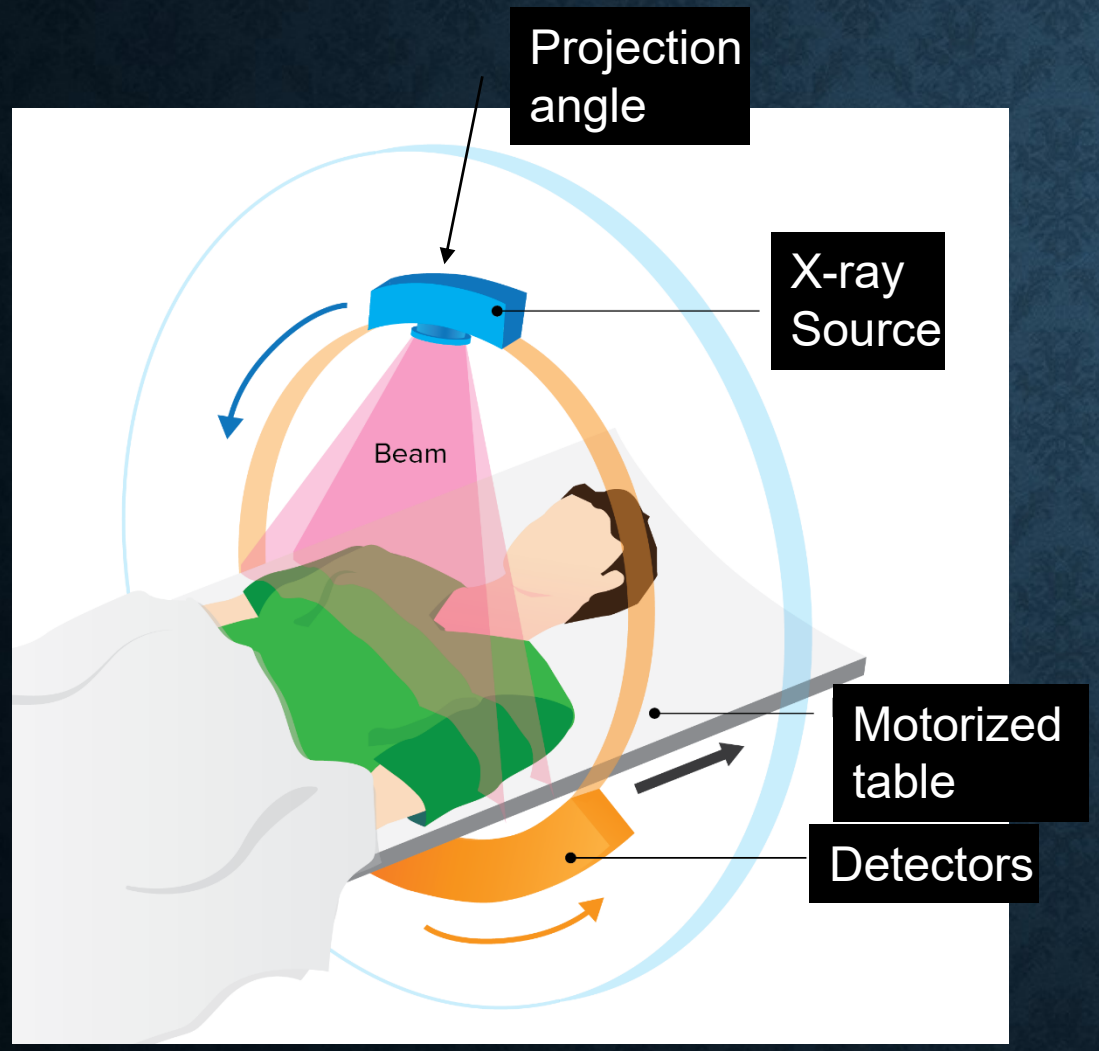
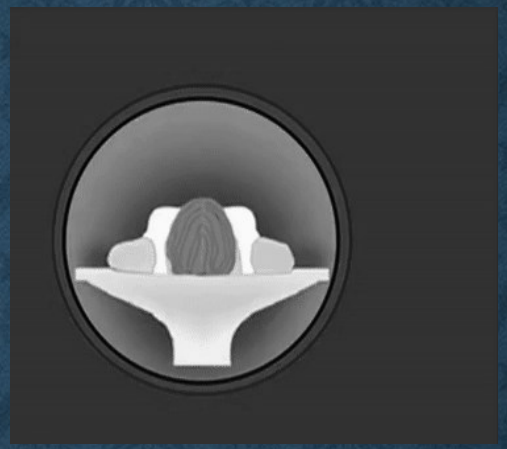
11 June 2025

OUTLINE

1. Sinograms-based Detection & Classification
2. Sino-CT Fusion
3. Cone Beam CT Reconstruction through Sinogram Enhancement
4. Domain Knowledge-based Augmentation for TBI Segmentation

Introduction to Computed Tomography (CT scan)

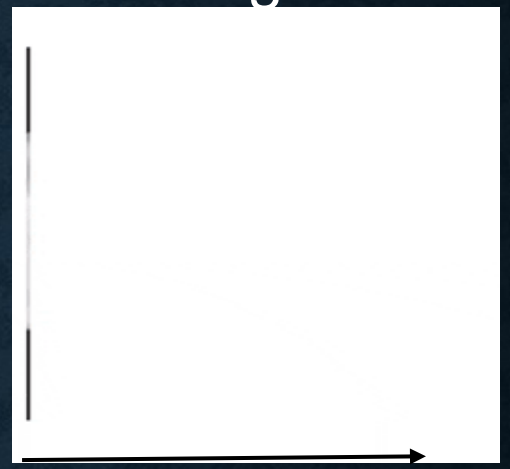




Milli second s (ms)

1

Sinogram



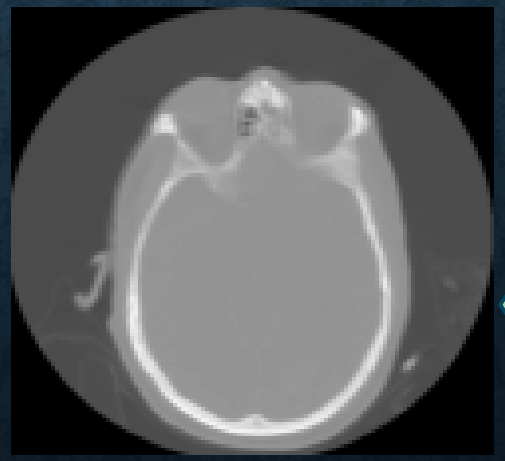
2

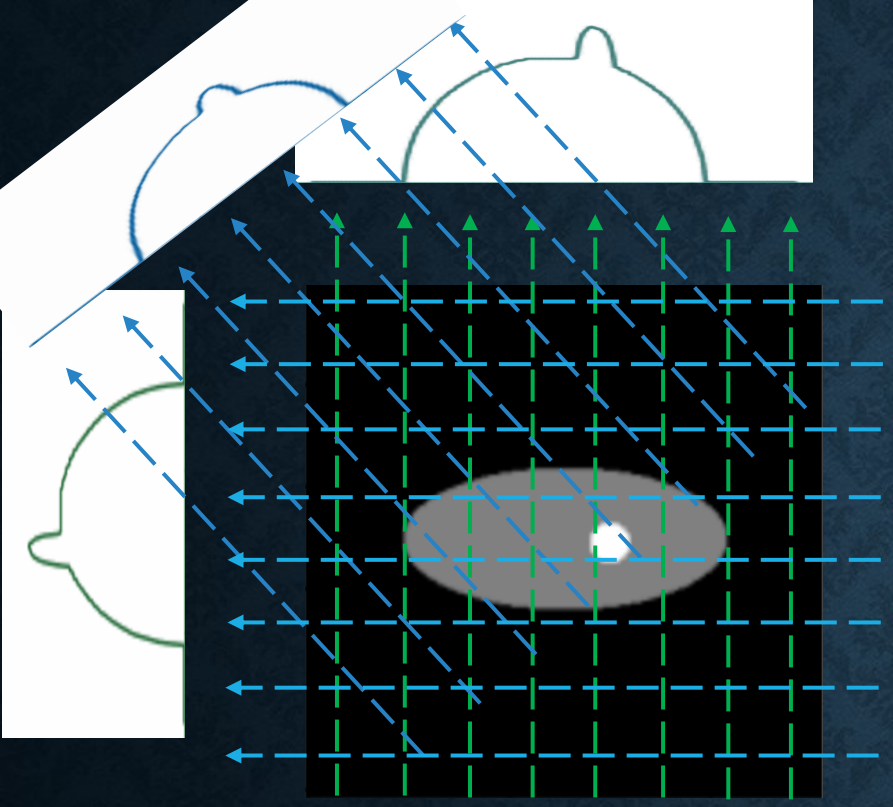
CT Reconstruction Algorithms

3

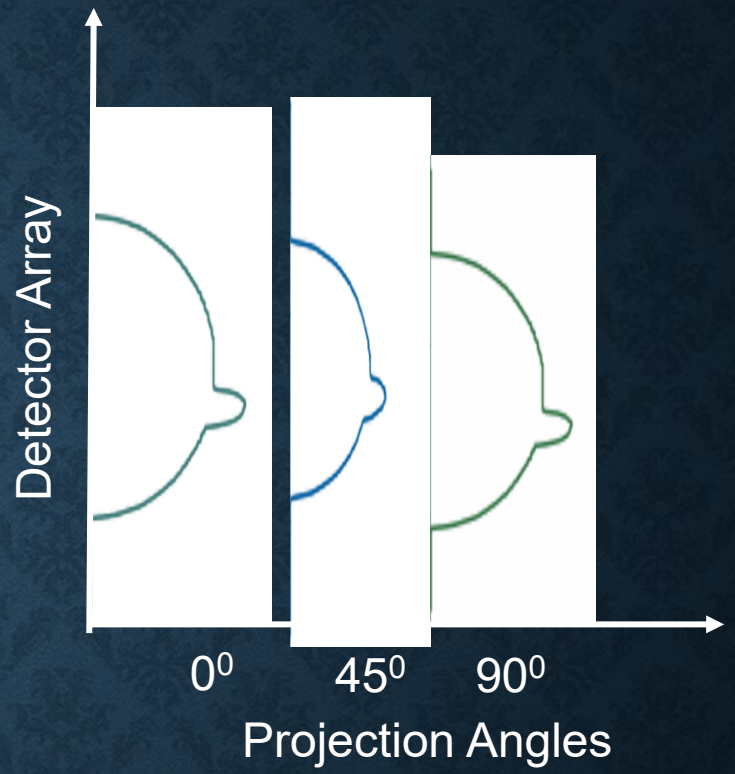
Up to 30 mins

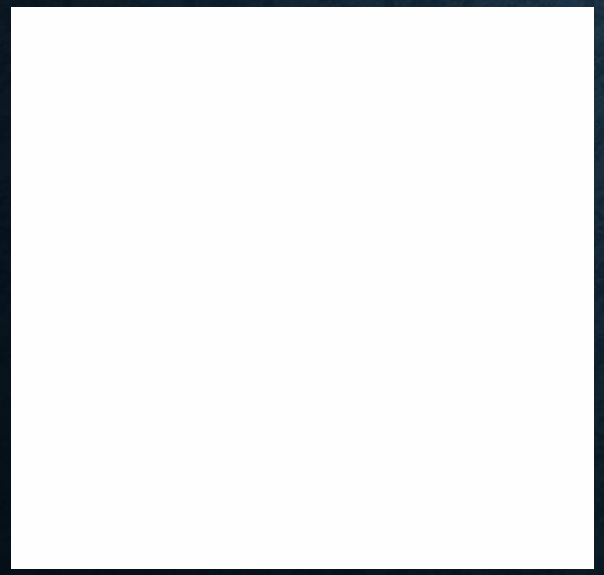
CT scan



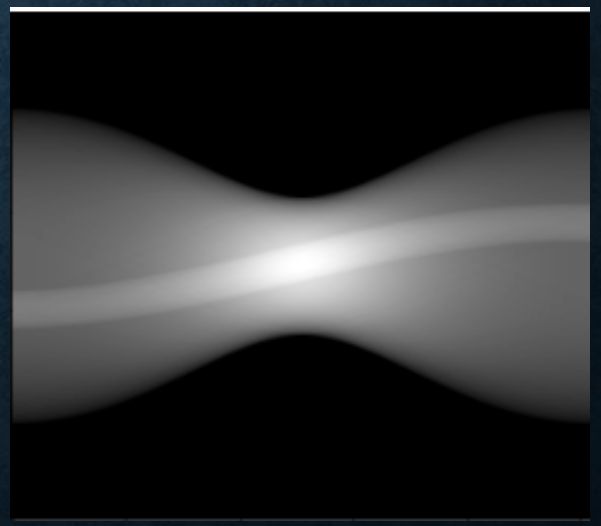


Sinogram acquisition



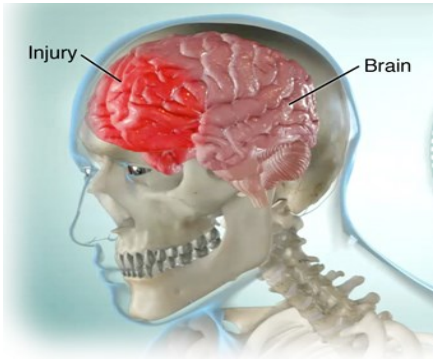


CT reconstruction



INTRODUCTION TO INTRACRANIAL HEMORRHAGES

What is ICH?



ICH occurs when a sudden trauma damages the brain, causing bleeding, bruising or tearing of nerves.

50 million Patients

Late & incorrect diagnosis

Delayed MRI & CT interpretation is one of the primary sources prohibiting the treatment.

Up to an Hour or beyond

Death & Disability

Without proper treatment, most of the ICH patients face death or permanent disability.

6 million deaths & disability

Socioeconomic burden

Permanent disability of the ICH patients has a huge social and economic impact on the society.

Costs over $US 400 billion

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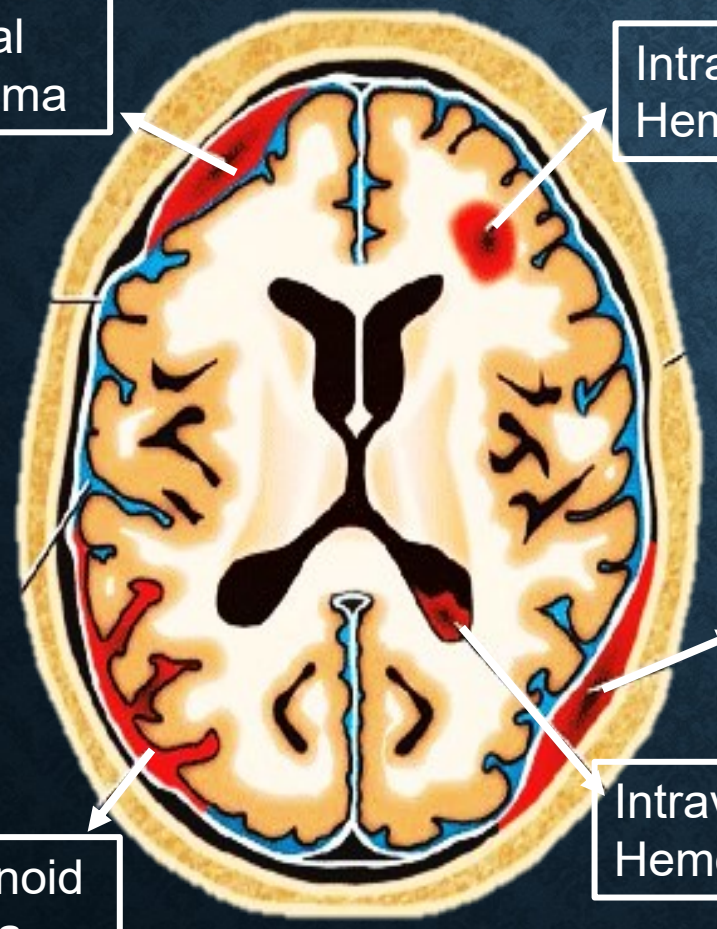
INTRODUCTION TO INTRACRANIAL HEMORRHAGES



CT is the choice of modality for ICH diagnosis

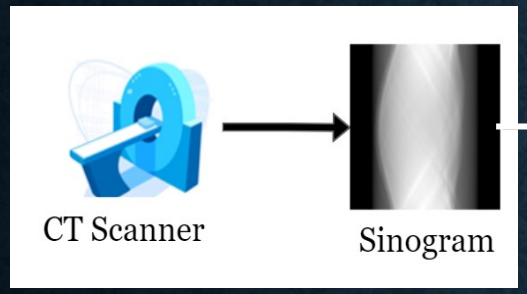
Subdural Hematoma

Subarachnoid Hematoma

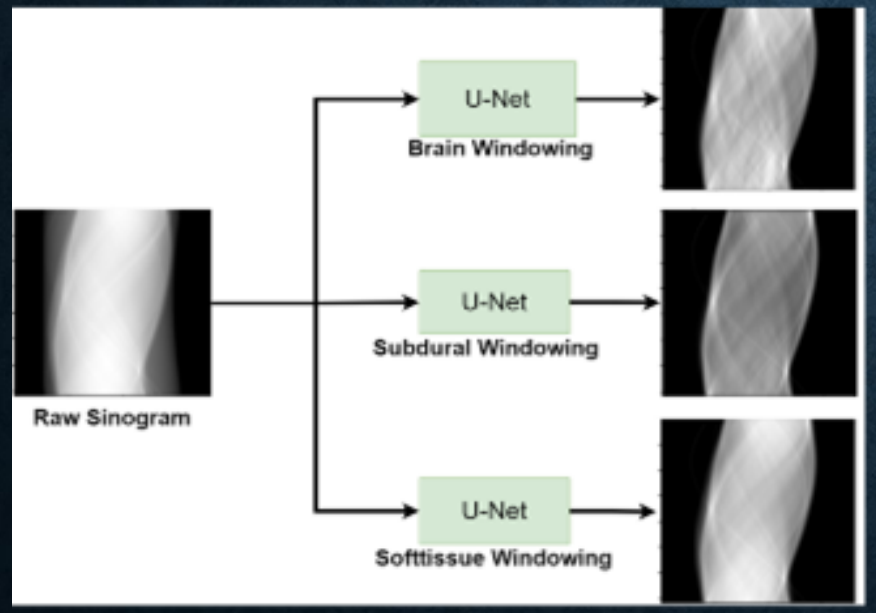
Epidural Hematoma

Intra cerebral Hematoma

Sinogram-based Method

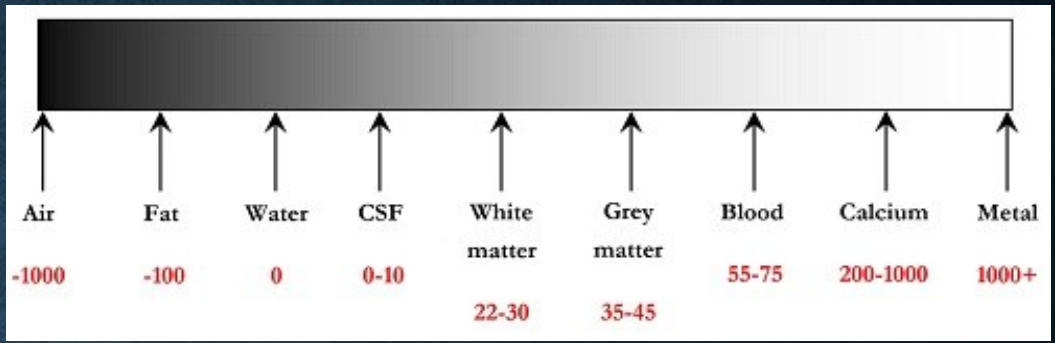


Intensity Transformed Sinogram Synthesizer Module(ITSS) → Detection Module → Detection of hemorrhage

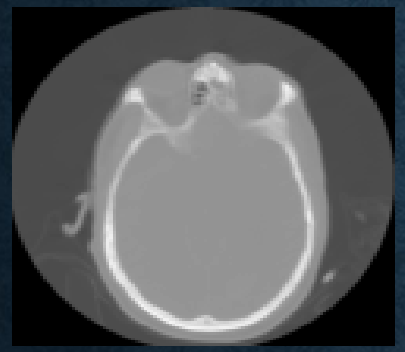


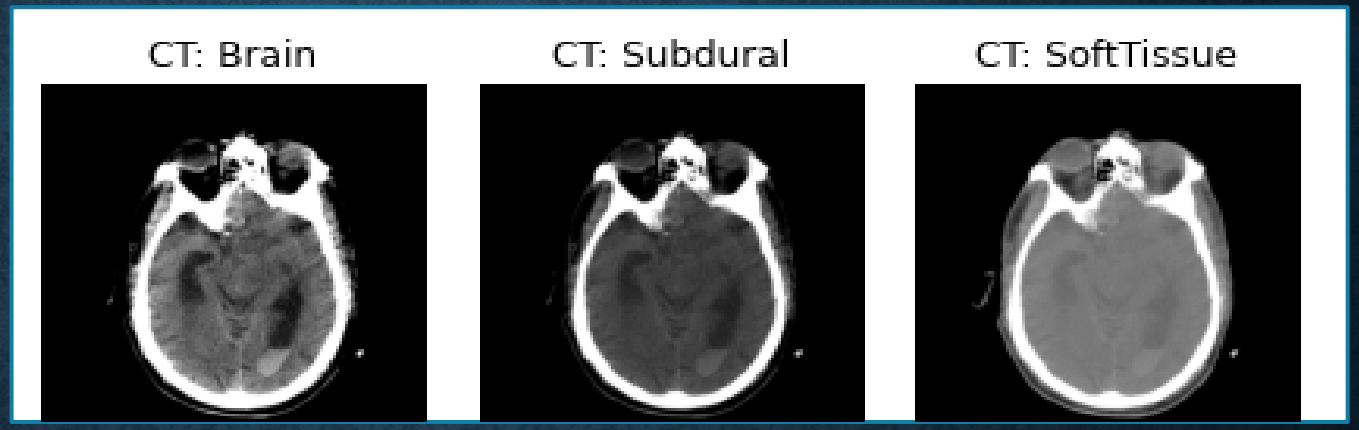
Convolution Neural Network + Recurrent Neural Network (CNN+RNN)

Sinogram-based Method



Raw CT

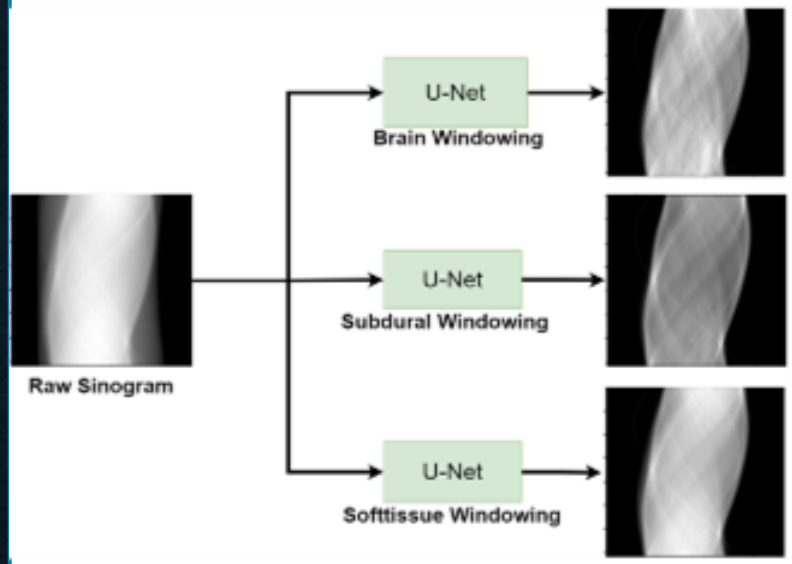


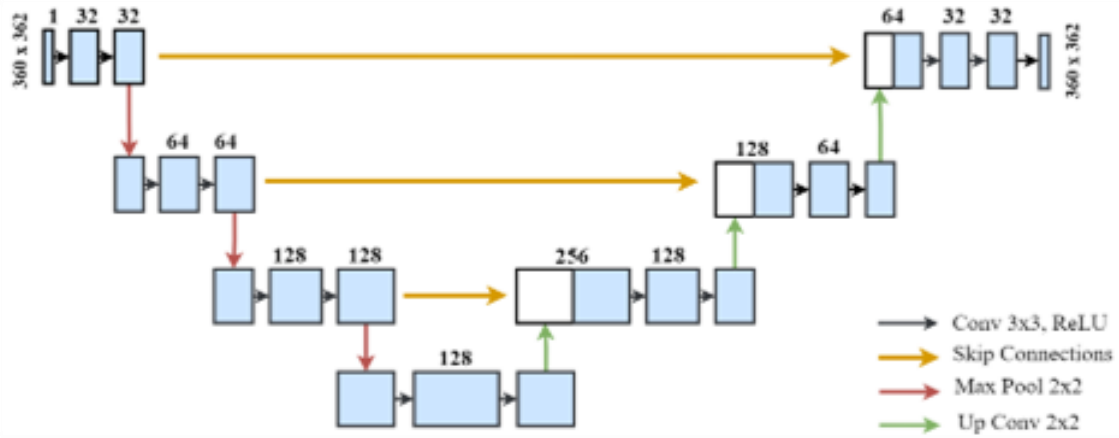


Raw sinogram

? ? ?

INTENSITY TRANSFORMED SINOGRAM SYNTHESIZER (ITSS)



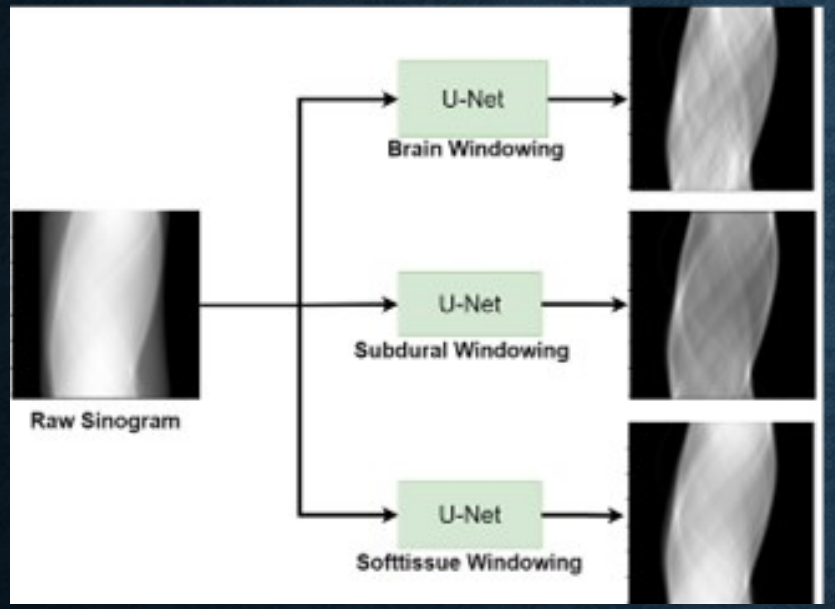


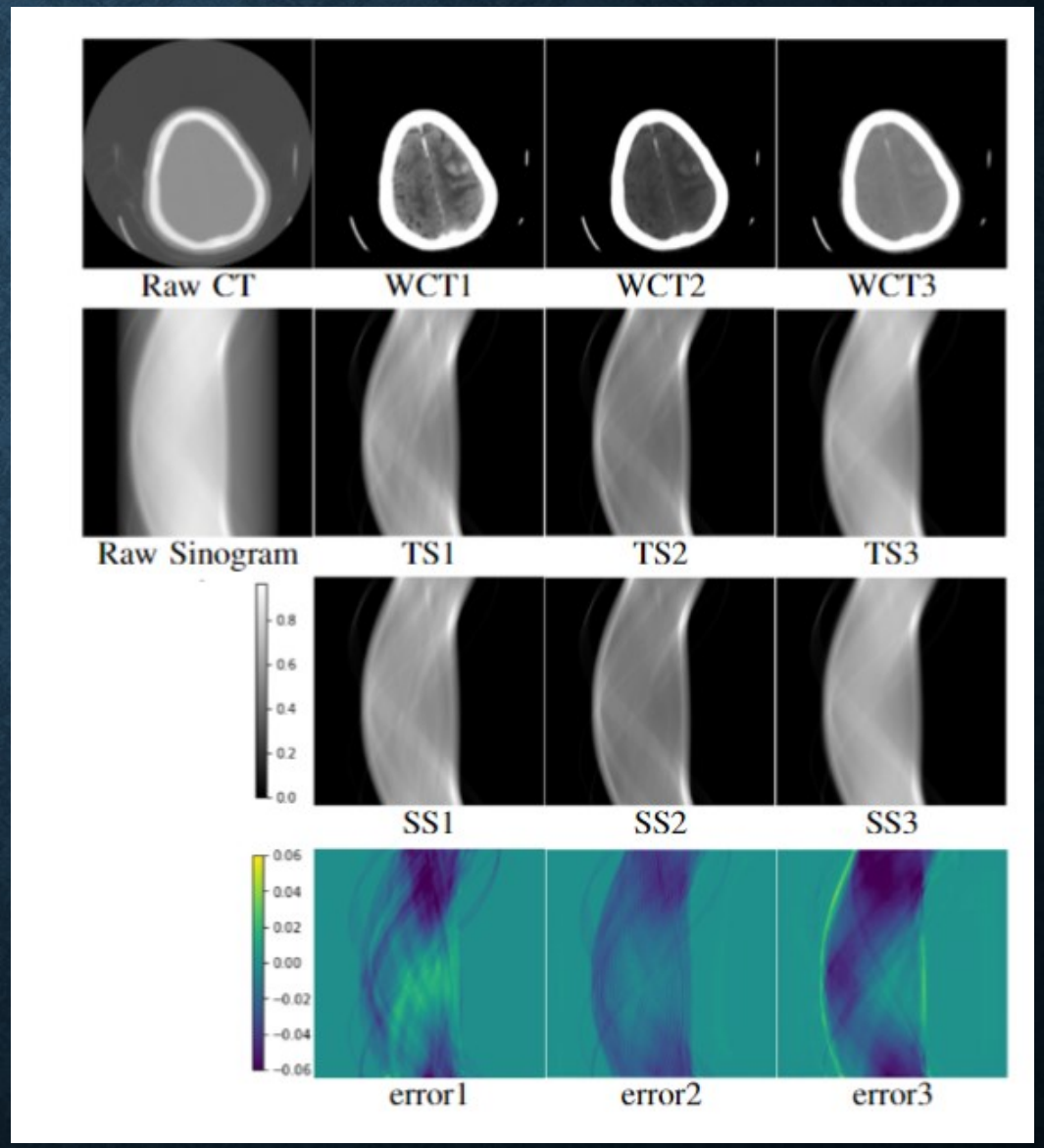
Quantitative evaluation of ITSS module on slices with and without hemorrhages for three windows

	Slices with hemorrhages		Slices without hemorrhages	
	MSE	MSSIM (%)	MSE	MSSIM (%)
Brain windowing	0.0067	95.53	0.0041	93.01
Subdural windowing	0.0221	90.36	0.0120	90.72
Soft tissue windowing	0.0023	96.80	0.0014	96.28

Sinogram-based Method

Intensity Transformed Sinogram Synthesizer (ITSS)





Sample result of ITSS module: Target Sinograms (TS), Synthesized Sinograms (SS) and the error are presented in rows one, two and three respectively

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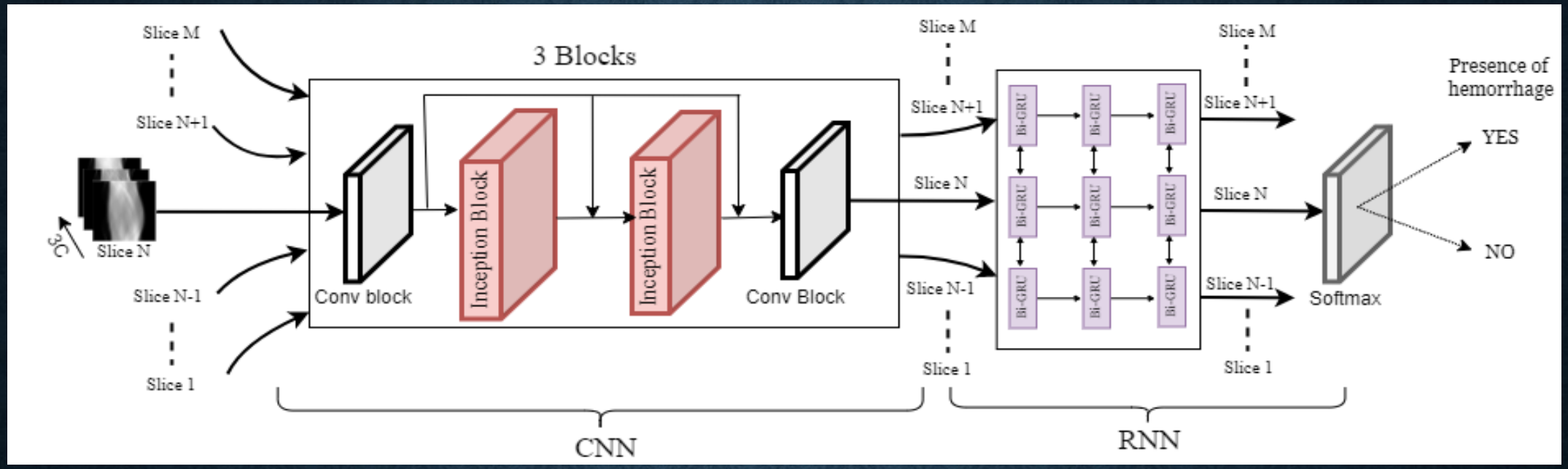
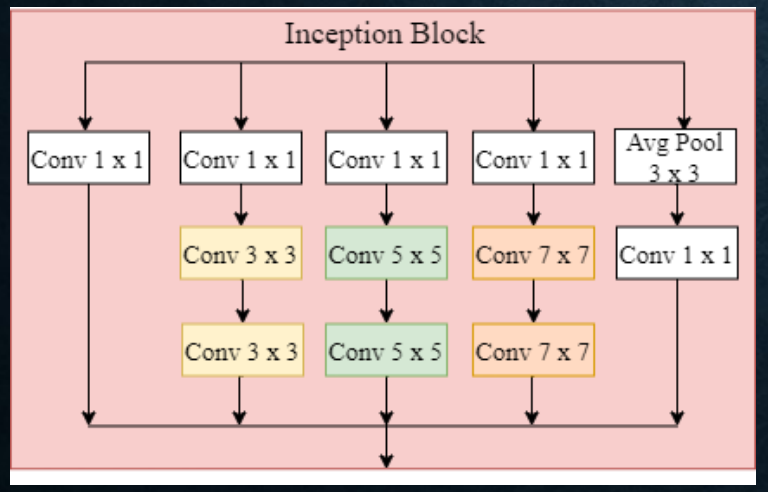
Detection Module (CNN+ RNN)

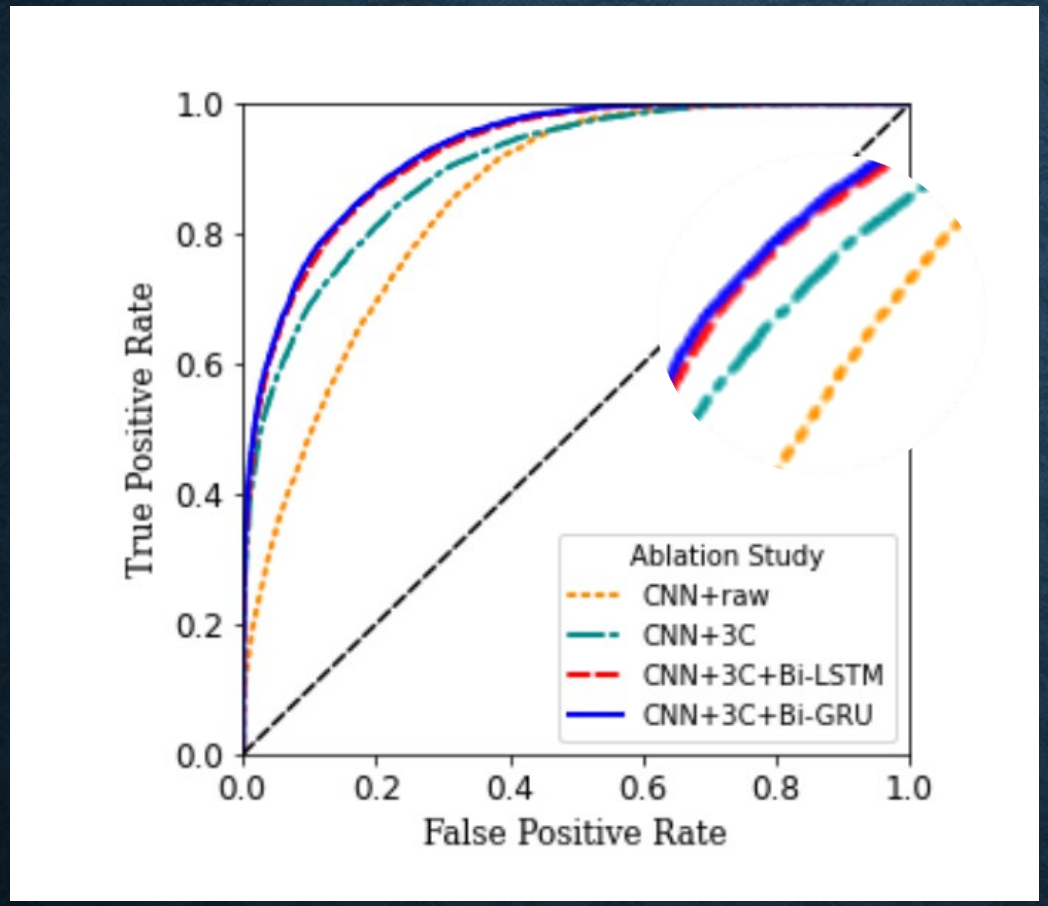




Quantitative evaluation of the proposed method with SOTA methods with the slice-wise (patient-wise) results

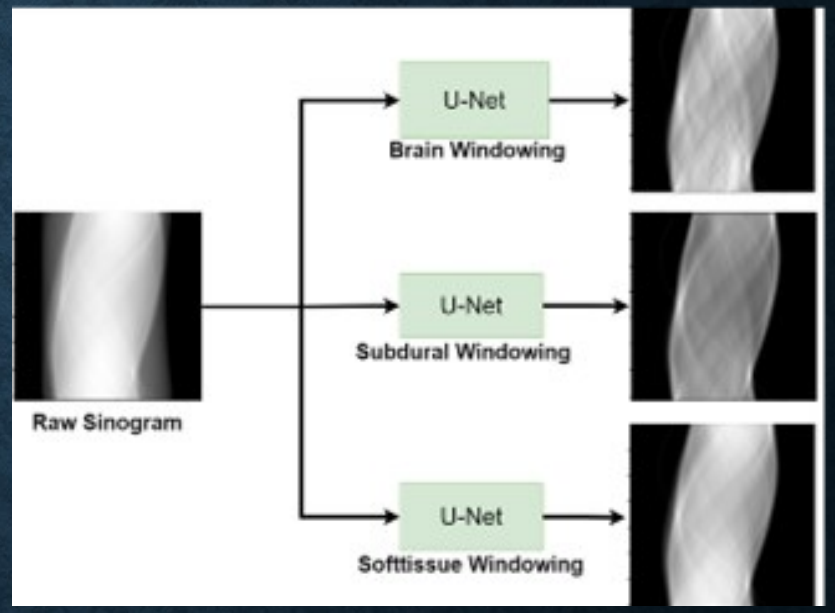
	Accuracy (%)	Sensitivity (%)	Specificity (%)
ResNeXt-101	55.73 (89.45)	**98.86** (70.87)	26.42 (92.64)
Inception-V3	68.70 (91.71)	98.14 (68.29)	48.69 (95.73)
ViT	57.22 (85.04)	94.71 (**98.18**)	26.90 (94.33)
Proposed Method	**95.50 (94.30)**	93.16 (87.81)	**97.10 (96.83)**

ABLATION STUDIES:

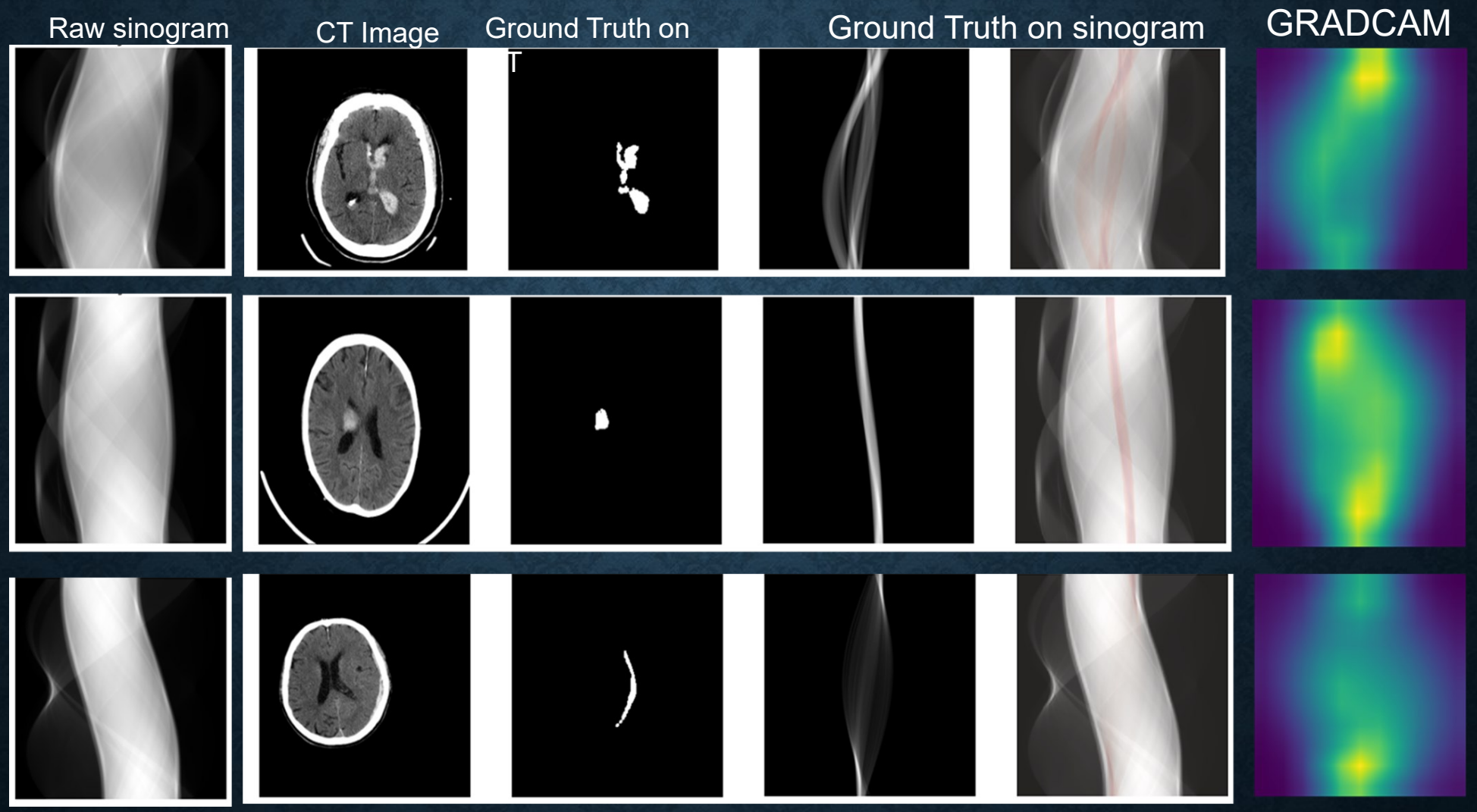


ROC curves for different number of channels (1C, 3C) and different RNNs (Bi-LSTM, Bi-GRU).

1C vs 3C



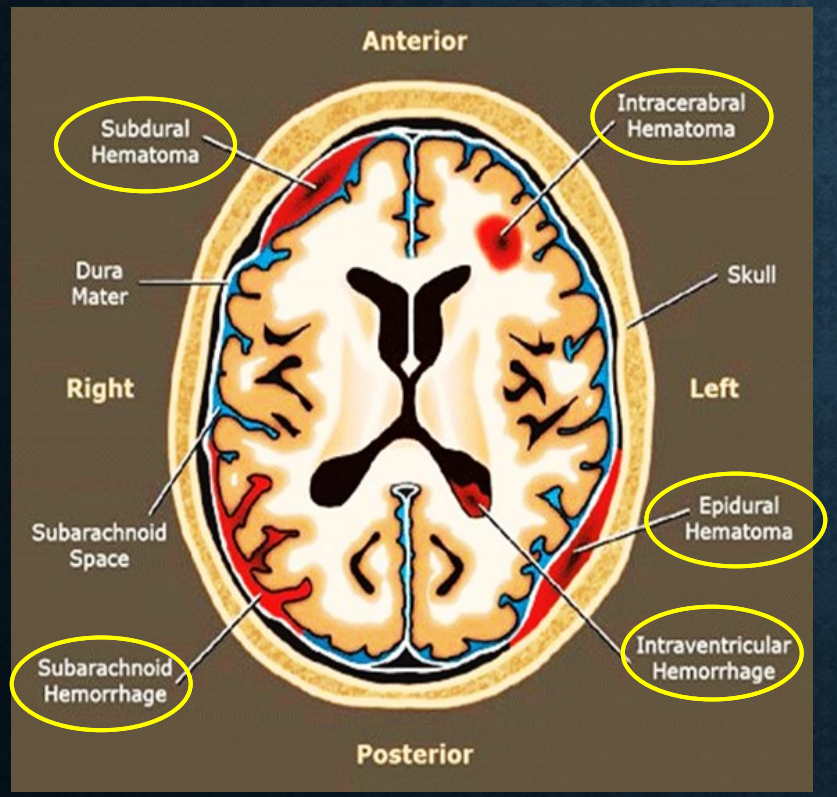
Interpretability through Grad-CAM heatmaps



GRADCAM results for three different types of injuries

Multi-label Classification

Different types of Traumatic Brain Injuries



Dataset Details

Class	#Patients
All	8726
EDH	170
IPH	2832
IVH	2165
SAH	2810
SDH	4017

RESULTS ON THE MULTI-LABEL CLASSIFICATION OF HEMORRHAGES

	Accuracy	AUC	Sensitivity	Specificity
EDH	99.45	55.73	11.76	99.70
IPH	96.34	65.49	31.40	99.62
IVH	96.44	78.12	58.36	97.88
SAH	85.75	74.11	61.24	86.97
SDH	88.67	78.21	66.11	90.31

The issue with the class imbalance in the dataset has been addressed in another work in the context of CT images

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CT-BASED DETECTION VS SINOGRAM-BASED DETECTION

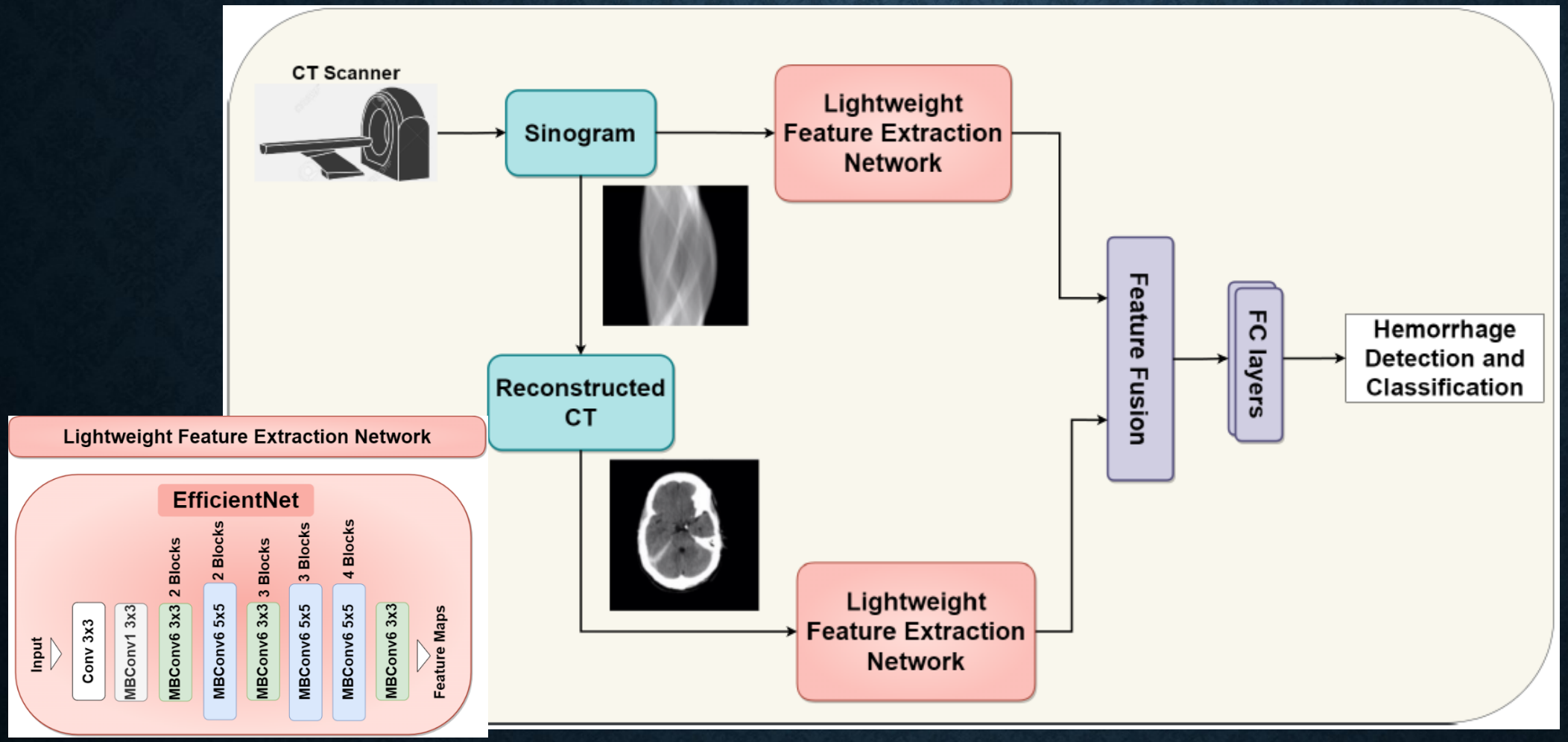
Robustness analysis of CT and sinogram-based detection to Poisson noise and offset errors in terms of accuracy

	CT-based Detection	Sinogram-based Detection
Initial Result	94.41%	94.30%
Offset Error	92.20% (2.21% ↓)	**94.15% (0.15% ↓)**
Poisson Noise	91.55% (2.86% ↓)	**94.20% (0.10% ↓)**

Introduced an offset error of 0.1° while acquiring the sinograms

Introduced an noise with Poisson distribution while acquiring the sinograms

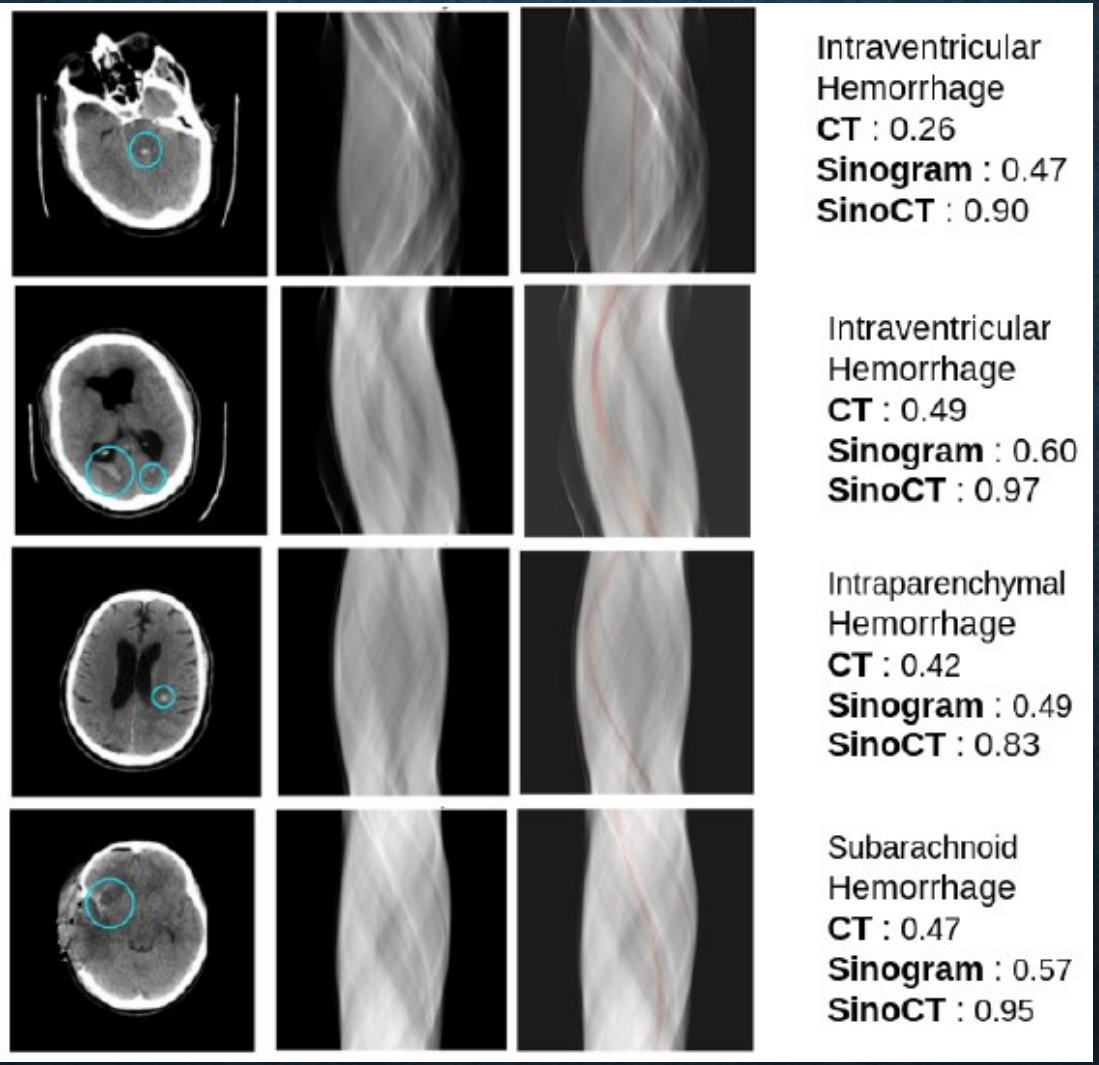
2. Sino-CT-Fusion for Detection & Classification of ICH



Sindhura

IEEE International Conference on Image Processing (ICIP), Abu Dhabi, UAE, 27-30 Oct 2024.

2. Sino-CT-Fusion for Detection & Classification of ICH



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2. SINO-CT-FUSION FOR DETECTION & CLASSIFICATION OF ICH

Table 1: Dataset Split for Training, Validation, and Testing.

Dataset	Subjects	Slices
Train	1000	42,724
Validation	200	8,421
Test	250	10,596
Total	**1450**	**61,741**

Table 2: Performance comparison of different input types on RSNA Dataset with two lightweight architectures.

Model	Input Type	Accuracy	AUC	Precision	Recall	No.of Parameters
EfficientNet	**CT**	81.34	81.76	**86.18**	51.17	4.41 M
	Sinogram	74.90	66.31	67.44	44.97	4.41 M
	CT + Sinogram	**84.07**	**84.27**	83.96	**63.40**	8.87 M
MobileNet V2	**CT**	74.70	**73.82**	**70.27**	45.19	3.47 M
	Sinogram	69.56	62.27	67.22	42.26	3.47 M
	CT + Sinogram	**76.68**	71.47	65.44	**60.80**	6.99 M

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3. ICASSP CHALLENGE PAPER (APR. 2024)

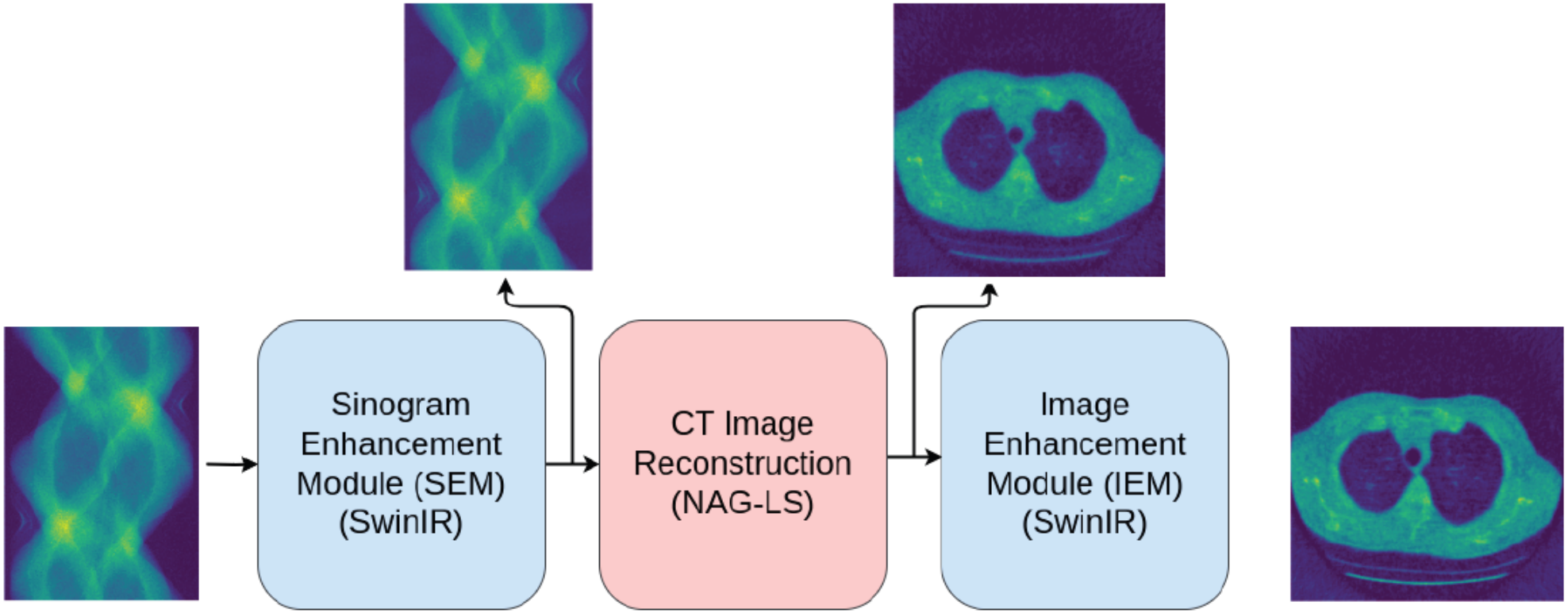


Fig. 1. Proposed CBCT reconstruction pipeline.

3. ICASSP CHALLENGE PAPER (APR. 2024)

Table 1. Average MSE for validation set.

Reconstruction Method	Low dose	Clinical dose
FDK (Baseline)	0.07959	0.03102
SIRT	0.06648	0.04545
NAG-LS	0.04408	0.04070
NAG-LS with SEM	0.01520	0.00940
NAG-LS with SEM and IEM	**0.00918**	**0.00467**

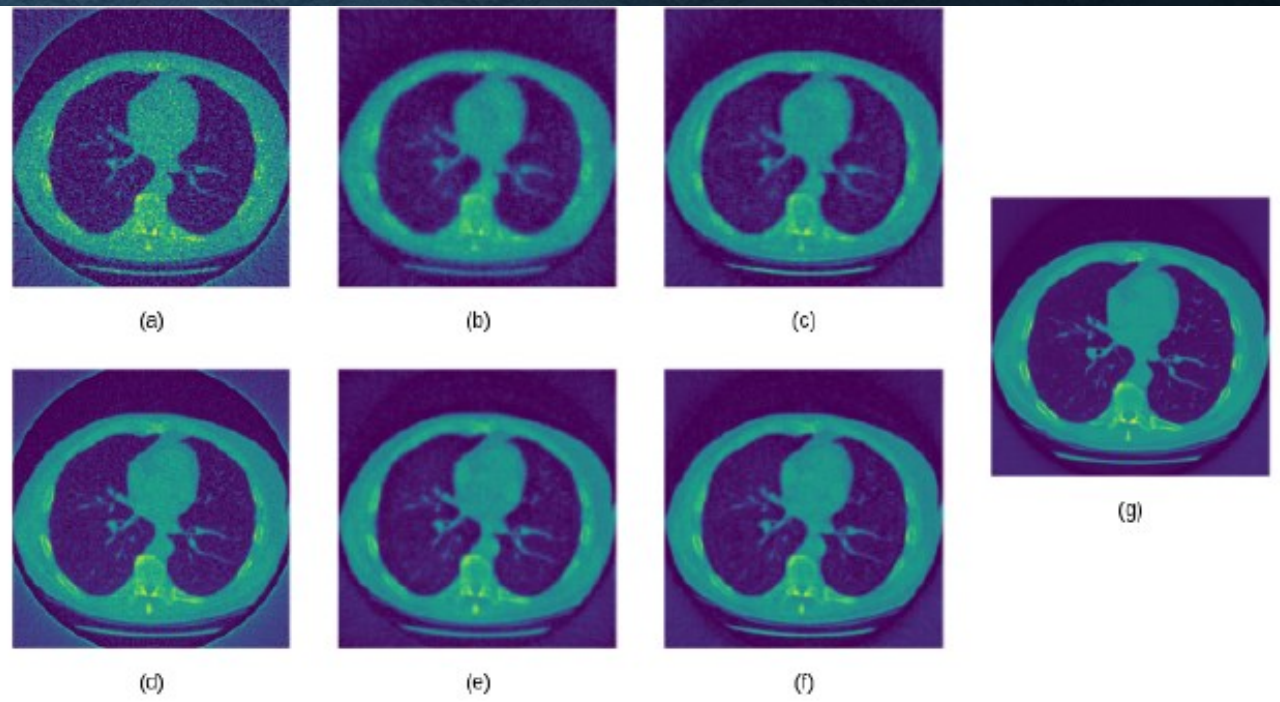
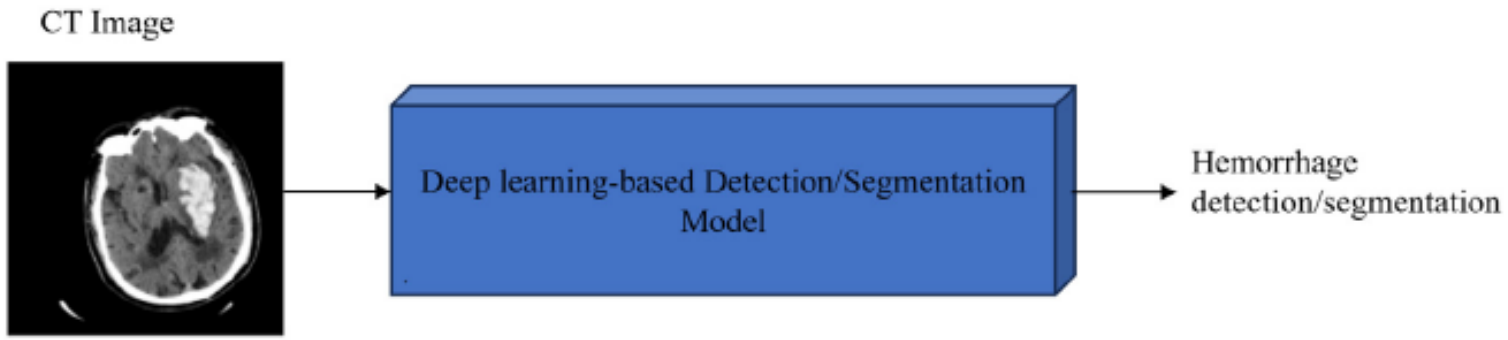
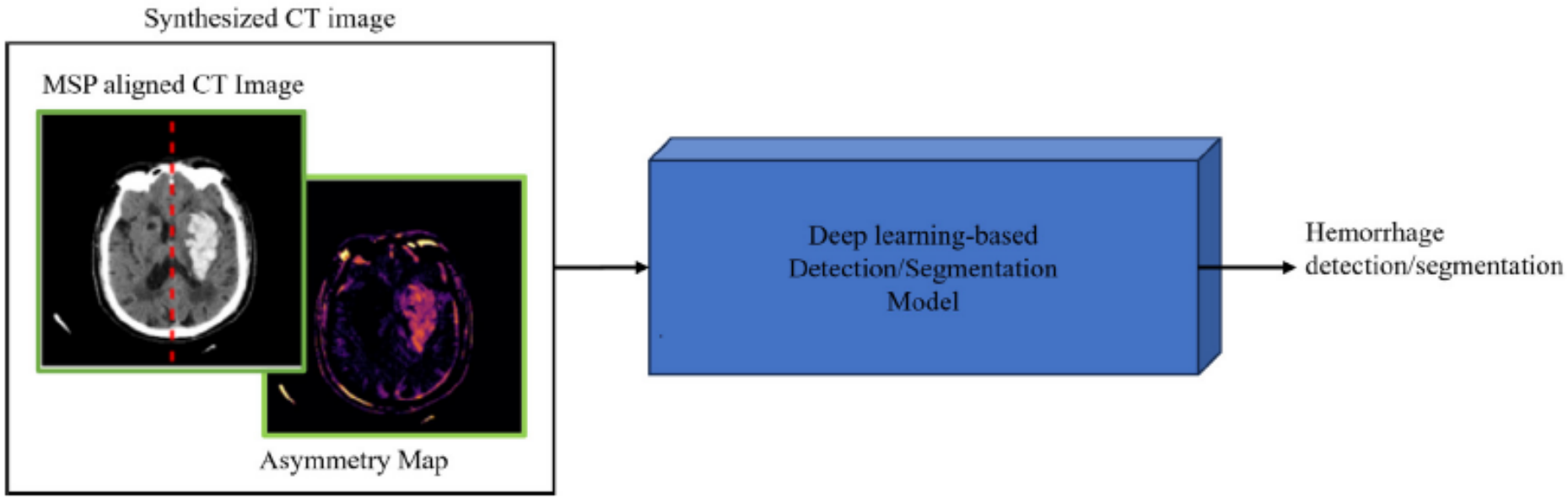


Fig. 3. (a), (d) are CT images reconstructed using FDK (Baseline). (b), (e) are CT images reconstructed using NAG-LS with SEM. (c), (f) are CT images reconstructed using NAG-LS with SEM and IEM. Upper and lower row images are low dose and clinical dose respectively. (g) is the clean CT image.

4. DOMAIN-KNOWLEDGE BASED DL SEGMENTATION



(a) A typical deep learning-based detection/segmentation framework.



(b) The proposed deep learning-based detection/segmentation framework.

Fig. 1. Proposed framework for detection and segmentation of hemorrhages from CT images.

4. DOMAIN-KNOWLEDGE BASED DL SEGMENTATION

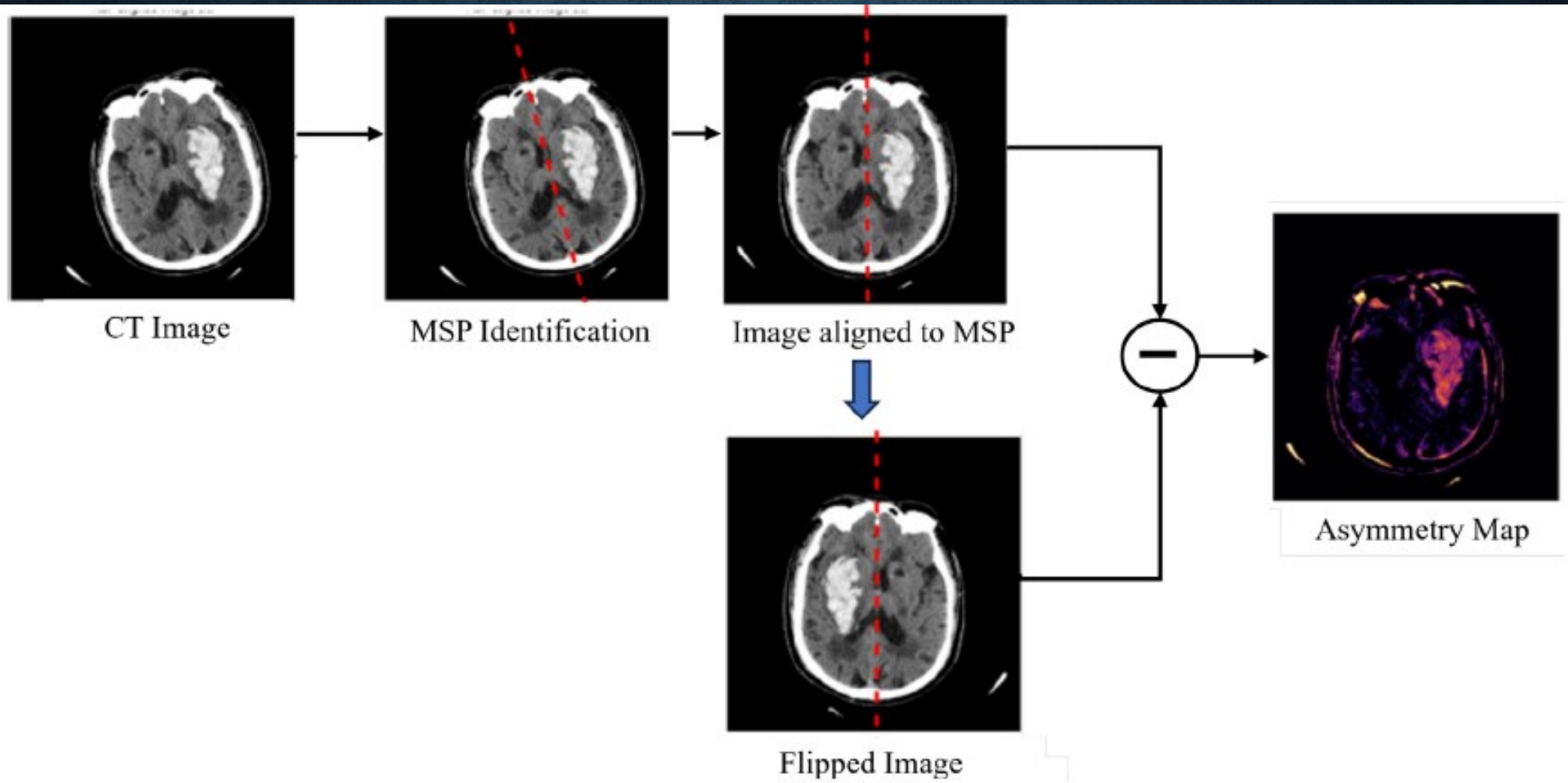


Fig. 2. Synthesizer Module: Workflow for generating an asymmetry map from a CT image.

4. DOMAIN-KNOWLEDGE BASED DL SEGMENTATION

Table 1
Comparison of segmentation models with and without using a synthesizer. Results are expressed as mean ± standard deviation, with all metrics presented in percentages.The symbol '×' indicates that the synthesizer was not used, while '✓' denotes that the synthesizer was used.

Model	Synthesizer	DSC (%)	IoU (%)	p-value
UNet	×	56.1 ± 8.6	46.6 ± 7.6	0.012
UNet	✓	66.8 ± 6.7	56.9 ± 6.4	
MulResUNet	×	57.0 ± 7.2	46.8 ± 7.1	0.006
MulResUNet	✓	64.9 ± 7.7	55.0 ± 6.8	
UNet3+	×	55.9 ± 7.7	45.9 ± 7.0	0.0007
UNet3+	✓	64.4 ± 8.3	54.3 ± 7.5	
U2Net	×	54.5 ± 6.2	44.7 ± 6.8	0.0038
U2Net	✓	62.2 ± 7.5	51.7 ± 7.8	

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4. DOMAIN-KNOWLEDGE BASED DL SEGMENTATION

Table 2
Comparison of detection models with and without using a synthesizer. Results are expressed as mean ± standard deviation, with all metrics presented in percentages. The symbol '×' indicates that the synthesizer was not used, while '✓' denotes that the synthesizer was used.

Model	Synthesizer	Accuracy	AUC	Precision	Recall	p-value
VGG-19	×	85.3 ± 5.0	91.2 ± 4.8	77.7 ± 5.2	70 ± 16.6	$<10^{-5}$
VGG-19	✓	89.1 ± 1.3	93.8 ± 0.9	85.4 ± 3.0	75.7 ± 4.8	
ResNet-50	×	63.9 ± 4.1	74.2 ± 3.2	58.1 ± 8.4	46.3 ± 8.6	$<10^{-5}$
ResNet-50	✓	72.5 ± 2.1	78.3 ± 2.5	55.7 ± 14.1	61.2 ± 11.1	
EfficientNet-B0	×	67.2 ± 5.4	71.7 ± 4.8	52.5 ± 10.2	48.8 ± 10.4	$<10^{-4}$
EfficientNet-B0	✓	70.8 ± 2.4	79.4 ± 3.2	58.7 ± 0.8	37.2 ± 33.6	

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QUESTIONS?